College of Industrial Technology

King Mongkut's University of Technology North Bangkok

เลขที่นั่งสอบ

Midterm Examination of Semester 1

Subject: 392153 Chemistry III

Date: 31 July 2013

Year: 2013

Section: 15 - 16

Time: 13.00-15.00

Name: ______________________ ID: ____________________ Class: ____________

Instructions:

1. Cheating will result in failure of all classes registered for the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbook, dictionaries and calculators are NOT allowed.
4. Electronic communication devices are NOT allowed in the examination room.
5. The examination has 10 pages (including this page), 3 sections and a total score of 70 points.
6. Write all your solution and answer on this examination sheet.

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Examination	Total score	Points
Part 1	15	
Part 2	10	
Part 3	45	
Total	70	

Answer sheet for Part 1

	a.	b.	c.	d.
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				

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Part 1: Multiple Choice (15 points)

Answers the question in the answer sheet.

1. Consider the reaction:

$$2\ C_3H_6 + 9\ O_2 \longrightarrow 6CO_2 + 6\ H_2O$$

If the rate of appearance of CO_2 is 0.40 mol/L/s, the rate of disappearance of O_2 is:

a) 0.10 mol/L/s

b) 0.90 mol/L/s

c) 0.60 mol/L/s

d) 0.30 mol/L/s

2. Which statement best explains why increasing concentration increases reaction rate?

a) The collisions become more effective.

b) The collision frequency increases.

c) The average kinetic energy increases.

d) The activation energy increases.

3. Which of the following is the correct set of units for the reaction rate, if concentration is measured in molarity and the time in seconds.

a) mol Ls

b) mol L^{-1} s^{-1}

c) mol^{-1} L^{-1} s^{-1}

d) none of these

4. Which three factors affect the rate of a chemical reaction?

a) temperature, pressure and humidity

b) temperature, reactant concentration and catalyst

c) temperature, reactant concentration and product concentration

d) temperature, product concentration and container volume

5. In the rate law, Rate = $k[NO]^2[O_2]$, the reaction is _____ order for NO, _____ order for O_2, and _____ order overall.

a) second; first; third

b) second; zero; third

c) first; second; third

d) first; third; first

6. From $2\ NO\ (g) + 2\ H_2\ (g) \longrightarrow N_2\ (g) + 2\ H_2O\ (g)$

Which of the following is true regarding the relative molar rates of disappearance of the reactants and the appearance of the products?

I. N_2 appears at the same rate that H_2 disappears.

II. H_2O appears at the same rate that NO disappears.

III. NO disappears at the same rate that H_2 disappears.

a) I and II only
b) I and III only
c) II and III only
d) I, II, and III

7. The reaction $A + B \longrightarrow C + D$ follows the rate law: Rate = k [A].

What will be the effect of decreasing the concentration of A?

a) The rate of the reaction will increase.
b) More D will form.
c) The rate of the reaction will decrease.
d) The reaction will shift to the left.

8. A reaction follows the rate law: Rate = $k[A]^2$. Which of the following plots will give a straight line?

a) 1/[A] versus 1/time
b) [A]2 versus time
c) 1/[A] versus time
d) ln[A] versus time

9. The half-life for the reaction at 550°C is 24 seconds. How many seconds are needed for the formic acid concentration to decrease by 33%?

a) 7.9 s   b) 14 s
c) 16 s   d) 17 s

10. Which one of the following is **NOT** a key concept of the collision theory:

a) particles must collide in order to react
b) particles must move slowly when they collide, otherwise they simply “bounce off” one another
c) particles must collide with the proper orientation
d) particles must collide with sufficient energy to reach the activated complex in order to react

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11. Which one of the following factors does ***not*** affect the rate of a chemical reaction:

   a) humidity

   b) concentration

   c) temperature

   d) nature of the reactants

12. Activation energy is the amount of energy required to.........

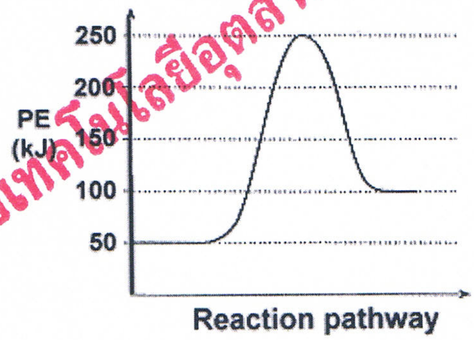
   a) break the bonds between the reactant and product molecules

   b) convert the reactants into the activated complex

   c) make the reacting particles collide

   d) form the bonds between the product molecules

Please use the following potential energy curve for questions 13 - 15.



13. The heat of reaction, ΔH, for the forward reaction is

   a) +200 kJ c) +150 kJ

   b) - 50 kJ d) + 50 kJ

14. The activation energy, E_a, for the reverse reaction is

   a) + 50 kJ c) -150 kJ

   b) + 200 kJ d) + 150 kJ

15. What is the type of the forward reaction?

   a) exothermic reaction c) endothermic reaction

   b) exothermic-endothermic reaction d) not enough information

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Part 2: True/False (10 points)

Indicate whether the sentence or statement is true (T) or false (F).

.......1. Rate is defined in terms of the change in concentration of a given reaction component per unit time.

.......2.The higher the activation energy barrier, the faster the reaction.

.......3. Increasing the concentration of a reactant may increase the rate of a reaction.

.......4. Adding a catalyst speeds up the rate of reaction for both the forward and reverse reactions.

.......5. Increasing the concentration increases the rate of a reaction, because it increases the number of collisions.

.......6. In general, as the temperature increases, the rate of a chemical reaction increases due a greater number of effective collisions.

.......7. For a first order reaction, the rate of the reaction doubles as the concentration of the reactant(s) doubles.

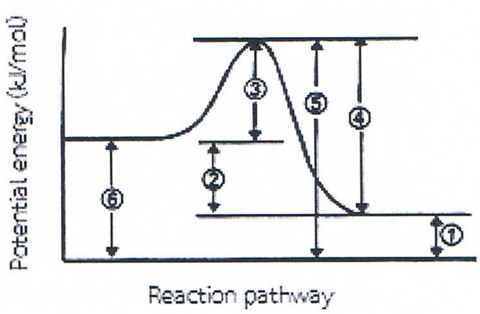
.......8. Catalyst makes reaction more exothermic.

.......9. The rate of a reaction usually changes if the concentration of the reactants is changed.

......10. Order is related to the coefficients in the balanced equation.

Part 3: Answer the questions (45 points)

1. Please use the number in the following potential energy curve to answer the following: (6 points)



a) Activation energy (Ea)............................

b) Heat of reaction (ΔH)...........................

c) Reactant Potential Energy.......................

d) Product Potential Energy ....

e) Potential Energy of the transition complex ....

f) Activation energy of reverse reaction...

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2. What are the relative rates of change in concentration of the products and reactant in the decomposition of nitrosyl chloride, NOCl? (3 points)

$$2\ NOCl\ (g) \longrightarrow 2\ NO(g) + Cl_2(g)$$

3. Ammonia can be formed by reacting nitrogen and hydrogen gases. (3 points)

$$N_2(g) + 3\ H_2(g) \longrightarrow 2\ NH_3(g)$$

If the rate of disappearance of hydrogen is -2.7×10^{-2} M/s, what is the rate of formation of ammonia?

4. Please fill the correct answer into a blank form. (9 points)

Order	Rate Law	Concentration-time equation	Half-life
0			
1			
2			

5. The half-life of a first-order decomposition reaction is 188 seconds. If the initial concentration of reactant is 0.524 M, what is the concentration of reactant after 752 seconds? (3 points)

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6. Answer the questions below based on the following reaction and initial rate data:

$$2A(g) + B(g) \longrightarrow C(g)$$

Experiment	[A]	[B]	Rate (M/s)
1	0.40	0.20	8.0×10^{-4}
2	0.80	0.40	1.6×10^{-3}
3	0.80	0.80	3.2×10^{-3}
4	0.60	0.60	2.4×10^{-3}

a) Write the rate law for the reaction above in the form Rate = $k[A]^x[B]^y$. (4 points)

b) What is the value of the rate constant for the reaction? (Be sure to include the proper units) (2 points)

c) Calculate the rate of formation of C in experiment 4 after [A] has decreased to 0.40 M. (2 points)

7. A first order reaction has a rate constant of 0.33 min^{-1}. How many minutes for the reactant concentration to decrease from 0.13M to 0.088M? (4 points)

8. Hydrogen peroxide decomposes according to the reaction $2H_2O_2(l) \longrightarrow 2H_2O(l) + O_2(g)$ which is first order for H_2O_2 and has a half life of 18.0 minutes. If an H_2O_2 solution that was initially 0.80M is allowed to decompose for 72 minutes, what will the concentration be at that time? (4 points)

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9. If a tree dies and the trunk remains undisturbed for 15,580 years, what percentage of the original carbon-14 is still present? The half-life of carbon-14 is 5730 years. (5 points)

Dr. Monrudee Phongaksorn

Dr. Sabaithip Tungkamani

...............................GOOD LUCK.................................